

Varun Mehrishi

Software Development Engineer II, Amazon | Distributed Systems, Payments and Audit Platforms

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PROFESSIONAL SUMMARY

Software engineer with 7 years at Amazon building event-driven microservices and REST APIs for financial systems that audit and pay carrier invoices across tens of billions of dollars of annual transportation spend. Designed and led a cross-team enrichment service, a dead-letter-queue platform adopted by six teams and a configurable dispute-detection engine; led five engineers and two interns to launch; reviewer of record for six payment and audit services where one defect costs millions. Seven years on-call, incident lead for Sev2 events. Java, AWS, Spark, Rust.

WORK EXPERIENCE

Software Development Engineer II, Amazon (Transportation Financial Systems), Hyderabad Jul 2022 – Present

- Designed the end-to-end architecture of Meghnad, a shared enrichment service filling missing shipment scan data from partner APIs and web sources for an org-wide audit-automation program; led three engineers to an Apr 2026 launch. Raised shipment scan-data coverage from **98% to 99.8%**; no DLQ or latency incidents since.
- Built its asynchronous pipeline (SNS/SQS, Parquet batches, cross-account S3, Spark on EMR) with distributed rate limiting to partner quotas, retries and health checks within a 2-hour SLA; shipped a decision-rollback control across three services in three days to unblock launch.
- Led Charge Type Payments to **100% carrier coverage**: owned readiness with four partner teams, grew EU from 15 to 130+ carriers and launched 230+ NA carriers with zero incidents; led two engineers to replace contention-prone optimistic locking with FIFO serialization by audit-item identity.
- Drove mitigation of an incident with **\$2.5M** in payments stuck, with the on-call senior engineer: sized the blast radius with a SQL query against the data warehouse, replaced last-write-wins amount merging with per-transaction summation, ran a staged replay under change management. Released in five days.
- Designed Configurable Property Dispute Detection (comparator abstraction, YAML model with polymorphic tags) so new dispute types ship as configuration; fixed a **\$17M** duplicate-resolution defect with an order-independent equivalence hash; cut DynamoDB storage cost **70%** via record compression.
- Reviewed design and code as engineer of record for six services paying carriers at multi-billion-dollar annual scale (**137 reviews in 2025**), blocking defects such as paid-amount aggregation before release; mentored two interns, one hired. Led incident recovery for a heap-exhaustion cascade and an EU surge (fleet 8 to 40 hosts); wrote recovery CLIs that restored 1,113 oversized records with zero failures.

Software Development Engineer, Amazon (Transportation Financial Systems) Nov 2019 – Jun 2022

- Authored TFS Helix, the dead-letter-queue platform (Java 11, jOOQ, Lambda, Multi-AZ MySQL with RDS Proxy, SQS, S3, CDK deployed through CI/CD pipelines) that prevents message loss, surfaces stack traces and redrives after fixes; adopted by **six teams** and still the shared platform in 2026.
- Built shared CDK alarm constructs rolled to four services; made Payment Authorization rules declarative through generic applicability criteria and deciders.
- Owned month-end close for auditing (2021 to 2022): resolved **35 finance-critical tickets** with zero escalations, reconciling payment data with SQL and Python; onboarded India carriers with 189 GST translation codes on the PostgreSQL-backed TIPS 1.0 invoice platform.

Microsoft Consulting Services, Associate Project Manager, Hyderabad

Jul 2019 – Nov 2019

PROJECTS

- rdp-parser (Rust, MIT): parser-combinator reader for the Redis RDB format with unit tests. Hackattic (Rust): ten systems challenges on one trait-based async runner (rayon, FFT, scrypt, WebSocket). github.com/varunmehrishi

SKILLS

Languages: Java, TypeScript, Scala, Python, Rust, SQL, Bash

AWS: Lambda, SQS, SNS, DynamoDB, S3, Kinesis Data Firehose, Step Functions, EMR, RDS (MySQL, PostgreSQL), ECR, API Gateway, CloudWatch, IAM, CDK v2, Athena, Redshift

Tools and practices: REST APIs, Docker, Smithy, Guice, Dagger, jOOQ, JUnit, Apache Spark, Elasticsearch, Node.js, React, CI/CD; system design, database design (DynamoDB, MySQL, PostgreSQL), message queues (SQS, SNS, Kinesis), caching, event-driven architecture, idempotency, observability, load and regression testing, on-call and incident management, code review, mentoring

EDUCATION

Manipal University Jaipur, B.Tech in Computer Science and Engineering, CGPA 9.36/10

2015 – 2019